

SeismoFNO: Physics-Grounded Neural Operators for Seismic Structural Response

A research brief on methodology, experiments, and open problems

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Why this problem

Simulating a multi-story building's response to an earthquake — tracking every floor's displacement over a 20-second shaking event — takes OpenSeesPy between 30 and 200 milliseconds per structure, depending on how much plastic yielding occurs. Regional seismic risk assessment involves tens of thousands of unique buildings across a city grid. Running high-fidelity nonlinear time-history analyses for each is prohibitive, which is why the field has historically relied on simplified empirical relations that discard most of the interesting physics.

Neural operators offer an alternative: learn the mapping from earthquake input to structural response once, then evaluate near-instantly. The governing equation is a coupled second-order matrix ODE:

$$M \ddot{u}(t) + C \dot{u}(t) + f_{\text{int}}(u(t), u(t)) = -M a_g(t)$$

From an operator-learning perspective this is a mapping between infinite-dimensional function spaces — ground acceleration and multi-story trajectory — conditioned on a structural parameter set that varies across buildings. Civil structures violate standard FNO assumptions: variable story count, variable interstory stiffness, and nonlinear hysteretic constitutive laws that dynamically shift natural frequencies during strong shaking.

1. Foundations: getting the physics right first (EXP1–EXP3)

Before training any surrogate model, three experiments established a trustworthy ground-truth pipeline. EXP1 validated single-degree-of-freedom elastoplastic dynamics and Bouc-Wen hysteretic loops against analytical solutions. EXP2 extended this to continuous hysteretic energy tracking. EXP3 designed the partitioning scheme — splitting by structural archetype *and* by earthquake identity simultaneously, so that each OOD partition has a clear, distinct physical meaning.

Partition	Structures	Earthquakes	n	Tests
In-distribution (ID)	5 known archetypes	RSN0001–08	120	Ordinary accuracy
OOD-A	5 known archetypes	RSN0011–12 (held-out)	300	Earthquake generalization
OOD-B	5S_T120 only (held-out, T1=1.20 s)	RSN0001–08	240	Structural generalization
OOD-C	5S_T120 (held-out)	RSN0011–12 (held-out)	60	Combined extrapolation

2. The zero-padding failure (EXP4)

A standard 2D Fourier Neural Operator on a fixed 5×2048 input tensor: five story slots, 1,024 time steps, with shorter buildings zero-padded. On five-story buildings: 19.29% median Relative L2 error. On three-story buildings: **99.60% median error**.

The cause is representational. A 2D Fourier transform assumes periodic boundary conditions. Zero-padded stories create a hard step from physical values to zero — a severe discontinuity that global spectral kernels spread as Gibbs ringing back down into the physical floors. No training duration fixes this because the representation itself is wrong for variable-height structures.

3. Graph topology eliminates the artifact (EXP5)

Replacing the fixed grid with a topology-native graph — nodes are floor slabs, edges are interstory columns, no padding — dropped 3-story median error from 99.60% to **22.09%**. Each operator block combines graph message-passing along structural edges with a 1D temporal Fourier convolution applied node-wise. 674,115 parameters, significantly fewer than the FNO2D (4,735,187).

A new failure appeared on the held-out flexible archetype 5S_T120 ($T_1 = 1.20$ s). The highest-period training structure is 5S_To85 at $T_1 = 0.85$ s, giving a 0.35-second extrapolation gap. On this unseen structure, EXP5 produced a **35.2% peak displacement error, 115.70% Relative L2 error**. Forensic FFT analysis confirmed the model was oscillating at ~ 1.12 Hz rather than the true 0.83 Hz. The unconditioned static Fourier kernels locked onto training frequencies; over 20 seconds, a 0.29 Hz frequency discrepancy drifts two signals to phase opposition, giving a mathematical relative error near $\sqrt{2} \approx 141\%$.

4. Modal conditioning via FiLM (EXP6)

Before any earthquake, the building's undamped natural frequencies are known from stiffness and mass matrices. Injecting these pre-earthquake modal invariants ($T_1, T_2, T_3, \omega_1, \omega_2, \omega_3$) into the operator via Feature-wise Linear Modulation:

$$[\gamma, \beta] = \text{MLP}(c), \quad h_{\text{mod}} = (1 + \gamma) \odot h + \beta$$

Model	Params	ID Peak Disp	OOD-A Peak	OOD-B Peak	OOD-C Peak
EXP4 FNO2D	4,735,187	—	51.74%	3.17% [†]	50.72%
EXP5 Baseline GNO	674,115	12.70%	15.86%	35.21%	38.66%
EXP6-B T1-GNO	725,059	2.09%	8.81%	13.47%	14.36%
EXP6-C Multi-modal GNO	727,619	8.87%	7.63%	13.06%	17.01%
EXP6-D Shuffled T1 (ablation)	725,059	2.65%	8.94%	24.33%	36.16%

[†]EXP4's OOD-B value is anomalous due to structural contamination in that split, corrected in EXP5/EXP6.

The ablation (EXP6-D) randomly permuted conditioning vectors across batch samples. OOD-B peak error rose from 13.47% to **24.33%**; OOD-C from 14.36% to **36.16%**. The gain is from physical modal correspondence, not auxiliary parameter capacity.

5. What still doesn't work

Relative L2 error on OOD-B remains above 100% with modal conditioning. Peak amplitude is controlled; waveform phase is not. Global 1D Fourier layers apply a static complex weight matrix over the entire 20.48-second record — they cannot adapt their frequency basis dynamically as the building softens during yielding. This is a fundamental architectural constraint, not a training failure. The natural successor is a causal state-space model (S4, Mamba) or neural ODE that processes time step-by-step.

6. Inference speed

Model	Batch	Latency	Throughput	vs. OpenSeesPy
OpenSeesPy 5-story NLTHA	1	54.68 ms	18.3 sim/s	1.00×
EXP4 FNO2D	1	6.60 ms	151.5 sim/s	8.28×
EXP5 Spatiotemporal GNO	1	16.25 ms	61.5 sim/s	3.36×
EXP6 T1-conditioned GNO	1	21.45 ms	46.6 sim/s	2.55×
EXP6 T1-conditioned GNO	32	30.19 ms	1,060 sim/s	~58× (throughput vs single sim)

Measured on Apple Silicon GPU (MPS) with process-wide device synchronisation. All latency values are medians over 50 runs.

7. Reproducibility

305 unit tests (pytest), zero-leakage partition verification, frozen checkpoints with SHA-256 hash verification, and an independent forensic audit script. All numbers in this brief are computed live from the evaluation CSVs at `results/experiments/exp6/evaluation/` and saved to `results/canonical/canonical_results.json`. No result is manually entered.

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